

LDO

- **Last Week:**
 - Met with the AnaTop team to discuss the enable signals.
- **Next:**
 - Implementing the layout for the enable signals.

LDO – Enable Signals

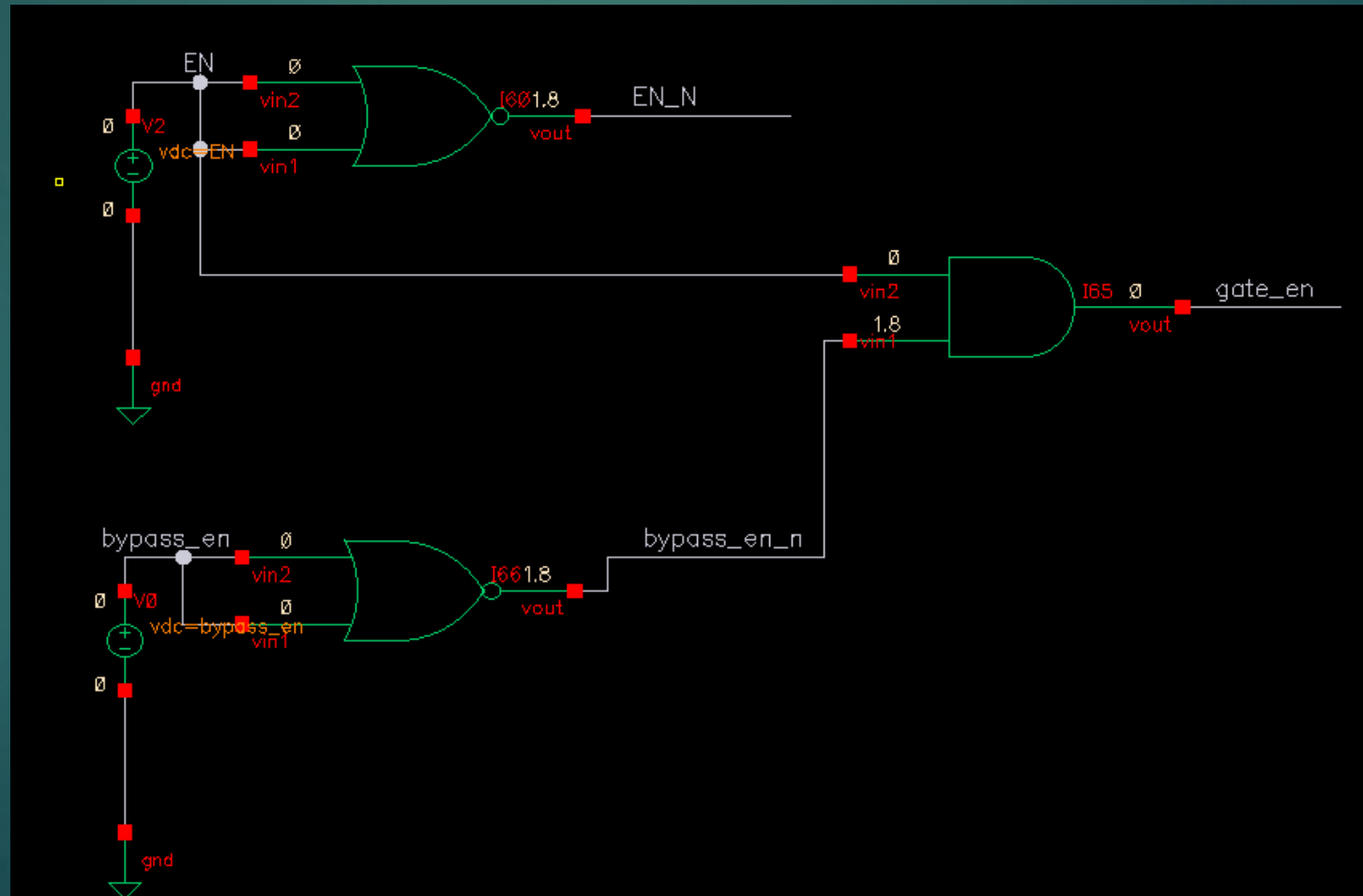
EN	bypass_en	Vout	gate_en = EN · bypass_en	Description
0	0	0V	0	NMOS pulls down output to 0V, PMOS pull-up disabled. Pass transistor disabled.
0	1	forbidden	0	Pulldown competes with pull-up, output undefined. Pass transistor disabled.
1	0	0.9V	1	Nominal operation. Pulldown and pull-up transistors disabled. Pass transistor enabled.
1	1	1.8V	0	PMOS pulls up output to 1.8V, NMOS pulldown disabled. Pass transistor disabled.

Table 1
Logic for Enable Signals

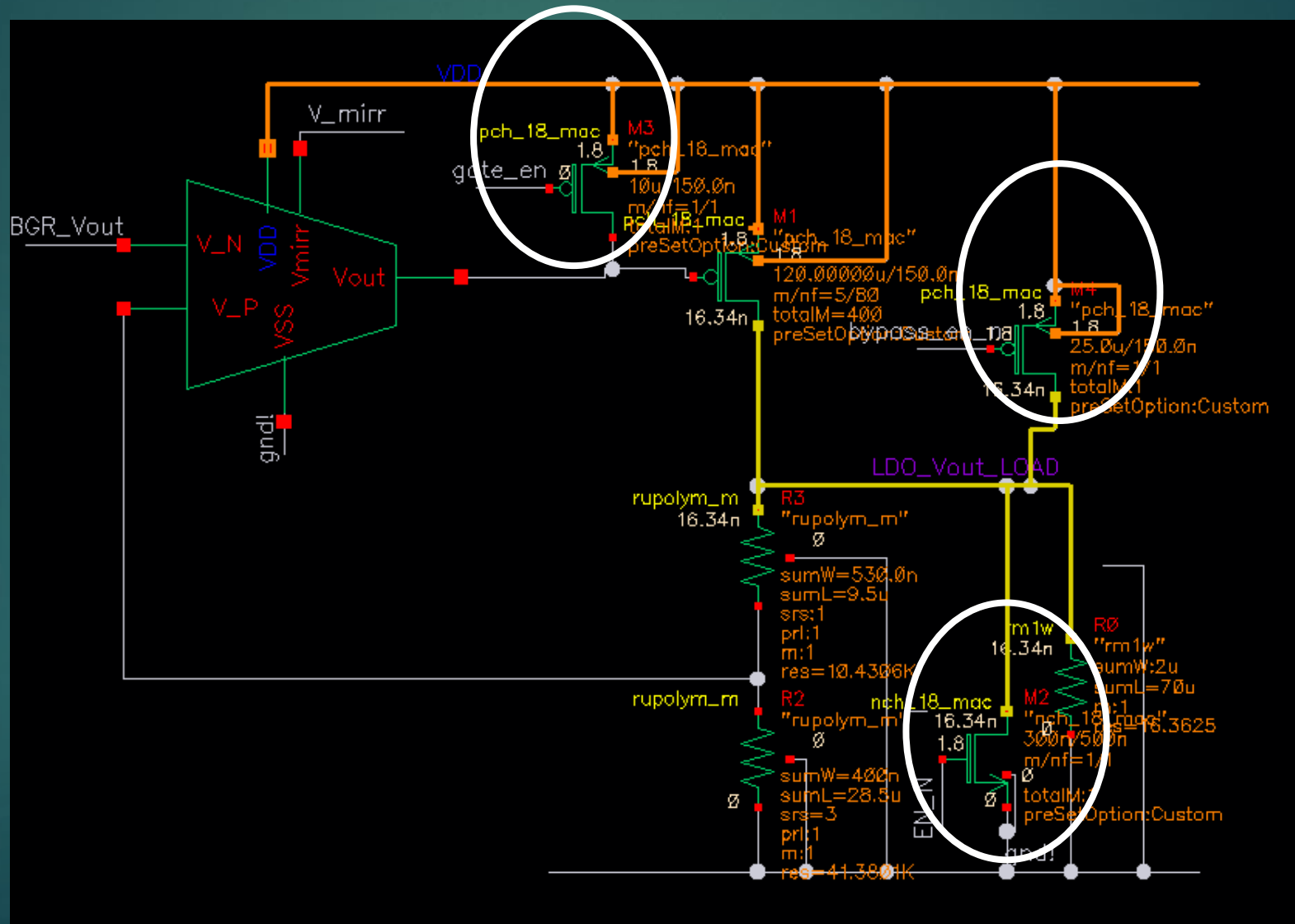
EN_N signal connected to pulldown NMOS gate
Bypass_en_n signal connected to pullup PMOS gate

LDO – Enable Signals

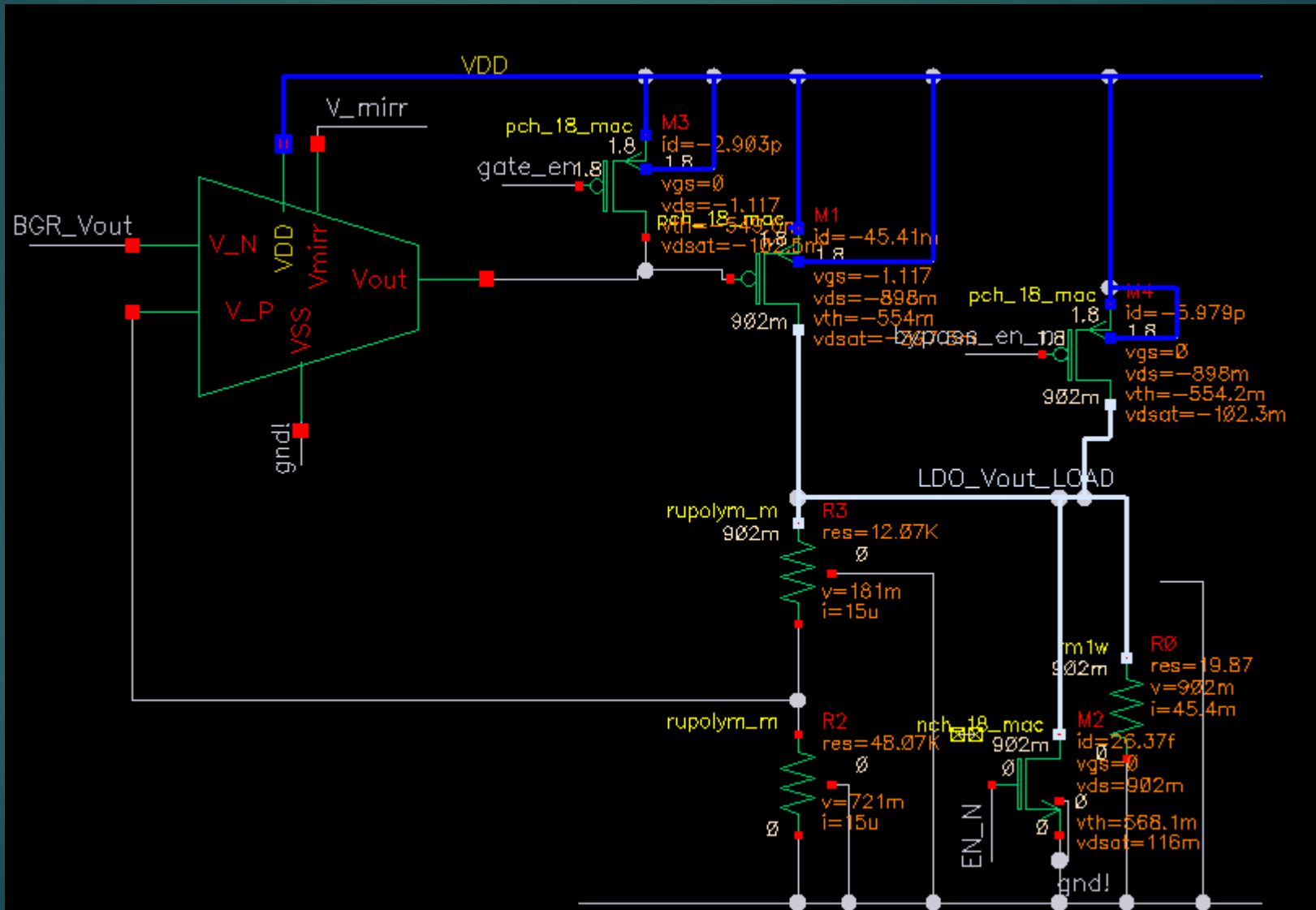
Logic gates should be from tcbn28 library



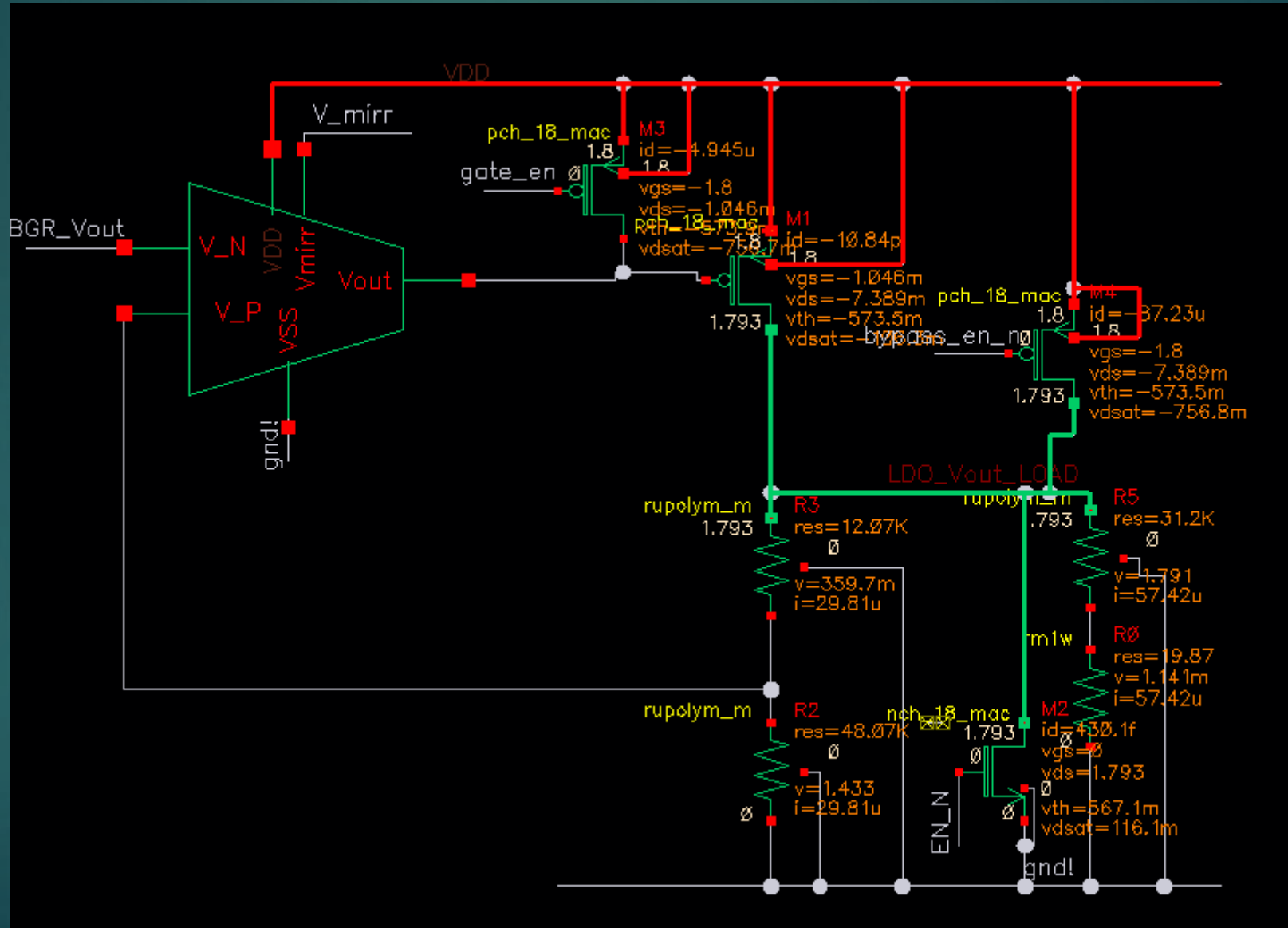
LDO – EN=0, bypass_en=0



LDO – EN=1, bypass_en=0



LDO – EN=1, bypass_en=1



LDO – EN=0, bypass_en=1

Depends if the PMOS pull up or the NMOS pull down is stronger.